Dense Supervised Dual-Aware Contrastive Learning for Airborne Laser Scanning Weakly Supervised Semantic Segmentation

Ziwei Luo, Tao Zeng, Xinyi Jiang, Qingyu Peng, Ying Ma, Zhong Xie, and Xiong Pan

***Abstract*— Airborne laser scanning (ALS) point clouds, characterized by low point density due to extensive ground coverage, pose significant challenges in labeling work. In addition, the ALS scanning pattern can lead to occlusions, resulting in data gaps and further increasing the difficulty of accurately labeling points within obstructed areas. In this context, fully supervised deep learning methods, which rely on a large volume of precisely annotated point-level data, are not well-suited due to the time-consuming and costly nature of obtaining extensive labeled datasets. Addressing these challenges, we propose an innovative weakly supervised learning strategy: a dense supervised dual-aware contrastive learning (DSDCL) approach. We introduce a context-aware contrastive constraint to enhance the model's ability to differentiate similar data points by leveraging subtle contextual differences, improving feature learning robustness in environments with incomplete objects. In addition, we apply a scene-aware dense contrastive constraint to enable multiscale point cloud representations while enforcing class-aware feature space constraints through gradient-based learning across multiple layers. We employ this contrastive constraint to enhance category discrepancies in prototype features predicted by pseudo-labels from dual-branch data, improving model robustness to categorical distribution differences by focusing on class prototype features and mitigating the impact of erroneous pseudo-labels. Extensive experiments on three benchmark ALS datasets demonstrate that DSDCL achieves performance comparable to advanced fully supervised methods, even when using only 0.1% of labeled samples.**

***Index Terms*— Airborne laser scanning (ALS) point clouds, contrastive learning, multiscale dense supervised, point cloud semantic segmentation, weakly supervised learning.**

Received 28 August 2024; revised 2 December 2024; accepted 23 January 2025. Date of publication 27 January 2025; date of current version 13 February 2025. This work was supported in part by Sichuan Science and Technology Program under Grant 2025ZNSFSC0529 and in part by the Natural Science Foundation of Hubei Province under Grant JCZRQN202500999. *(Corresponding author: Xiong Pan.)*

Ziwei Luo and Xiong Pan are with the School of Computer Science and Artificial Intelligence, Wuhan Textile University, Wuhan 430200, China (e-mail: luoziwei@cug.edu.cn; pxjlh@163.com).

Tao Zeng is with the School of Electronic Information Engineering, Sichuan University, Chengdu 610065, China, and also with Chengdu Qianjia Technology Company, Chengdu 610207, China (e-mail: sczt21@163.com).

Xinyi Jiang and Zhong Xie are with the School of Computer Science, China University of Geosciences (Wuhan), Wuhan 430074, China (e-mail: Xinyi_Jiang@cug.edu.cn; xiezhong@cug.edu.cn).

Qingyu Peng is with the State Key Laboratory of Intelligent Manufacturing Equipment and Technology, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: pqyhust@hust.edu.cn).

Ying Ma is with the National Engineering Research Center for Geographic Information System, China University of Geosciences (Wuhan), Wuhan 430074, China (e-mail: maying2019@cug.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3535104

I. Introduction

AIRBORNE laser scanning (ALS) point clouds provide accurate 3-D representations of large-scale urban scenes. A key step in fully interpreting complex geographical scenarios is the acquisition of semantic information, which serves as valuable cues in a variety of remote sensing applications including forest monitoring [1], [2], land cover surveys [3], [4], 3-D mapping [5], [6], and change detection [7], [8]. Semantic segmentation of ALS point clouds assigns a label with a semantic category to each point in urban scenes, providing a crucial solution for scene analysis and understanding.

The field of point cloud semantic segmentation has undergone significant evolution [9], [10], [11], [12], propelled by deep learning breakthroughs. This evolution is largely attributed to the development of sophisticated network architectures, resulting in a series of methodologies that have recurrently broken new ground [13], [14]. Most studies focus on designing new network architectures or convolutional kernels [15], [16], [17], [18], disregarding the significant expenses associated with ensuring the availability of a large number of accurate point-level labels. However, these methods are contingent on a vast amount of point-level annotation data, which is labor-intensive and time-consuming. The process of data labeling is not only laborious but also complex, especially for experienced personnel [19]. Collecting accurate annotation labels typically requires a substantial amount of labor, often demanding that professional operators meticulously complete the assignment. In addition, labeling ALS point clouds typically requires operators to confirm the category of points from multiple angles, making the annotation task particularly challenging [20]. Moreover, occlusions due to the scanning patterns of ALS systems result in data gaps, further complicating the determination of precise labels for points in obscured areas. If similar performance can be obtained with fewer labels, it would dramatically reduce the workload and significantly enhance the practicality of applications.

Recently, weakly supervised learning has emerged as a powerful tool for decreasing labor-intensive efforts in 3-D semantic segmentation of point clouds. Several methods have been developed to differentiate points across various scenes by leveraging the consistency of features at different levels under diverse data augmentations. For instance, exploring color and geometric smoothness [21], [22], as well as more advanced consistency losses such as Jensen–Shannon divergence [23] and similarity-weighted loss [24], to enhance

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

feature consistency between point clouds subjected to different augmentations or geometric calibrations [25], [26], [27].

Although these weakly supervised methods using consistency constraints have achieved certain results, there are still some limitations that need to be addressed. In ALS datasets, certain objects such as utility poles and bridges are relatively sparse points and corresponding few point-level labels, leading to a class imbalance issue. Under such circumstances, point-level consistency constraints may result in suboptimal learning performance for the network with respect to these classes. In the case of limited annotations, only exploring feature consistency would be insufficient to capture the complex structures of point clouds, making it difficult to extract both object and contextual information for satisfactory segmentation performance. Meanwhile, these methods have not fully considered the semantic characteristics of neighbors in scenes and the global features of 3-D classes, failing to fully utilize the limited but valuable annotations. In addition, these methods only operate on a single scale to directly regularize the feature space at the final layer of the encoder, but it's crucial for segmentation to recognize and delineate objects at multiple scales.

Contrastive learning has achieved strong results in shaping the feature space of networks under the constraints of the scarce samples [28], while avoiding representation collapse [29], establishing that contrastive pretraining on global [30] or dense [31] features can make the encoder comparable to supervised encoders when fine-tuning for downstream tasks. Based on the above analysis, we employ dense supervised dual-aware contrastive learning (DSDCL) to fully utilize sparse annotations by considering semantic neighborhood relations in scenes and the global features of 3-D scenes. We propose a context-aware contrastive constraint to enhance the model's ability to differentiate similar data points by leveraging subtle contextual differences, thereby improving feature learning robustness in environments with incomplete objects. In addition, we introduce a scene-aware dense contrastive constraint that enforces class-aware feature space constraints to enhance category discrepancies in prototype features from dual-branch pseudo-labels. This improves the model's robustness to categorical distribution differences and enables it to capture more complex relationships between classes in 3-D scenes. Furthermore, we apply class-aware constraints to the feature space throughout various stages of the model. Intuitively, we extend these contrastive constraints across multiple layers of the model, distributing the class-aware feature space constraints across different stages. For airborne point cloud data covering extensive areas, this framework accurately infers the semantic information of ground objects from sparse point cloud annotations. Extensive experiments on three benchmark ALS datasets demonstrate that DSDCL achieves performance comparable to state-of-the-art fully supervised methods, even when using only 0.1% of labeled samples. In conclusion, the primary contributions of this work are as follows.

1) We propose a context-aware contrastive constraint to enhance the model's ability to differentiate similar data points by leveraging subtle contextual differences, improving feature learning robustness in environments with incomplete objects.
2) We propose a scene-aware dense contrastive constraint that enforces class-aware feature space constraints through gradient-based learning across multiple layers. This contrastive constraint enhances category discrepancies in prototype features from dual-branch pseudo-labels, improving model robustness to categorical distribution differences and mitigating the impact of erroneous labels.
3) We propose a weakly supervised framework DSDCL to leverage specific contrastive properties to augment consistency constraints for ALS point cloud semantic segmentation. Our DSDCL's effectiveness is validated through extensive experiments, showcasing its superior performance compared to contemporary weakly supervised methods. Remarkably, it even exceeds the capabilities of numerous fully supervised learning networks.

The subsequent of this article is organized as follows. Section II provides a detailed review of current deep learning models for point cloud semantic segmentation. Section III provides a comprehensive overview of DSDCL. Section IV presents extensive experimental results of the semantic segmentation. Section V contains discussions on these results, and Section VI concludes with potential directions for future research.

II. Related Works

Deep learning techniques have achieved remarkable accuracy across numerous tasks. However, their dependency on meticulously annotated data limits real-world applicability. Addressing this, weakly supervised approaches aim to sustain high performance with fewer labels. This section offers an extensive review of research in 3-D weakly supervised areas. This approach is still in the early stages of development compared to fully supervised methods. Current techniques in weakly supervised learning, faced with incomplete and sparse annotations, integrate extra constraints as supplementary information during the training process. These methods, which enhance the learning capabilities of networks, employ diverse strategies and workflows.

A. Similarity Supervision Methods

The principle of similarity supervision maintains that visual interpretations or computed features should remain consistent when analyzing pairs of similar data. Consistency constraint, a widely used type of supervised constraint, involves applying modifications like scaling and rotation to the original point cloud. It aims to minimize the prediction differences between the two data branches as much as possible. PSD [23] incorporates multibranch point-level consistency constraints to exhibit significant performance. Wei et al. [24] developed a repositioning module that improves features across samples and introduced consistency loss through comparison with basic features. Xu and Lee [22] developed a novel multibranch supervision strategy, that includes simple point cloud transformations and a graph-based consistency framework based on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

the Laplacian matrix. Nonetheless, for large-scale point clouds with sparse labels, using a complex Laplacian matrix is less practical.

B. Pseudo-Labeling Methods

Wang and Yao [32] efficiently generate supplementary supervisory sources online using a nonparametric method through a soft pseudo-labeling approach. To enhance feature learning from imprecise pseudo labels, Lin et al. [33] utilized a supervised contrastive loss that emphasizes the inherent relationships among class-specific features. Wang et al. [34] formulated a context-aware pseudo-labeling strategy with a quasi-scene-level labeling approach to enhance point-level surveillance signals. SemanticFlow [35] incorporated a bi-directional cross-frame pseudo-tag propagation module that leverages scene flow to learn relationships and propagate pseudo-tags between neighboring frames. It also featured a tag refinement mechanism aimed at selecting reliable pseudo-tags for effective learning. Su et al. [36] utilized multi-stage scene-level constraints to represent unlabeled data using pseudo-labels. In addition, they developed an adaptive weighting strategy guided by uncertainty to minimize the adverse effects of erroneous pseudo-labeling.

C. Contrastive Pretraining Methods

Contrastive constraints supervise from the opposite direction of consistency constraints, seeking to draw anchor points nearer to augmented data points and distance them from other points within the prediction space. Xie et al. [27] first introduced a pretrained framework that employs contrastive learning to extract robust feature representations from unlabelled point cloud data, significantly improving performance across tasks within an unsupervised learning method. Jiang et al. [37] integrated pseudo-label guidance and confidence into the contrastive loss, using the guiding effect of both to avoid negative pairings within the same category. Hou et al. [25] utilized contrastive scene contexts to reduce the reliance on extensive labeled datasets. HybridCR [26] employed hybrid contrastive regularization that integrates local and global contrastive constraints to enhance learning from limited labeled data. These regularizers promoted similarity between anchor points and their corresponding positive points, while fostering dissimilarity from negative points regarding posterior probability.

III. Methodology

We begin by introducing the basic concepts and definitions of weakly supervised learning. Following this, we provide a brief overview of the DSDCL framework, including its core architecture and primary functionalities. Finally, we delve into a detailed examination of the key components of the framework, which include the hybrid point cloud augmenter, the point-level consistency constraint, the context-aware contrastive constraint, and the scene-aware dense contrastive constraint.

A. Preliminaries

In the context of weakly supervised segmentation tasks, our training dataset consists of two parts: labeled and unlabeled data points. Specifically, the dataset can be represented as $\{(P_l, Y_l), (P_u, \varnothing)\}$, where P_l denotes the set of M labeled points $(p_l^1, y_l^1), \ldots, (p_l^M, y_l^M)$, and P_u represents the set of unlabeled points $(p_u^{M+1}), \ldots, (p_u, ^N)$. The goal is to develop a function $f: P \rightarrow Y$ that assigns accurate semantic labels to each point in the dataset. The dataset P is a combination of both labeled and unlabeled points $P = P_l \cup P_u$, but only a small fraction of these points are labeled.

For a more precise setting, for the 1% configuration, the number of labeled points M is equal to 1% of the total points N; similarly, in the 0.1% setting, M is just 0.1% of N. In addition, the feature set of P is denoted as $F \in R^{(N \times d)}$, with d denoting the number of feature channels. The term $X \in R^{(N \times 3)}$ is used to describe the 3-D coordinates of the points in P, respectively.

B. Overall Framework

For limited labeled data and a large amount of unlabeled data, our DSDCL framework focuses on learning useful representations from a mass of unlabeled data to enhance the model's generalization ability. The overall framework of DSDCL is shown in Fig. 1. Initially, the raw point cloud and the augmented point clouds, generated by applying multiple transformations through various data augmenters, are fed into the network, which shares parameters between the two branches. Subsequently, we employ a standard cross-entropy loss $\mathcal{L}_{\text{seg}}$ and a consistency constraint $\mathcal{L}_{\text{con}}$, to maintain prediction consistency between the original and augmented branches as the basic losses. Given the limited number of labeled data, DSDCL incorporates two contrastive constraints to encourage the network to acquire more discriminative features for segmentation. Specifically, on one hand, DSDCL introduces a context-aware contrastive constraint $\mathcal{L}_{\text{CD_cra}}$ for direct feature-space supervision. This constraint enhances the model's ability to differentiate similar data points by leveraging subtle contextual differences, thereby improving feature learning robustness in environments with incomplete objects. In addition, it enables the model to grasp more intricate relationships between regions. On the other hand, DSDCL employs a scene-aware dense contrastive constraint $\mathcal{L}_{\text{SD_cra}}$ to enforce class-aware feature space constraints through gradient-based learning across multiple layers. This contrastive constraint enhances category discrepancies in prototype features derived from pseudo-labels, improving model robustness to categorical distribution differences and mitigating the impact of erroneous labels

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{con}} + \lambda_1 \mathcal{L}_{\text{CD_cra}} + \lambda_2 \mathcal{L}_{\text{SD_cra}}. \quad (1)$$

These constraints used in DSDCL will be discussed in the subsequent sections: a point-level consistency constraint (see Section III-D), a context-aware contrastive constraint (see Section III-E), and a scene-aware dense contrastive constraint (see Section III-F).

C. Hybrid Point Cloud Augmentation

Data augmentation employing global similarity transformations such as translation, scaling, and rotation, along with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

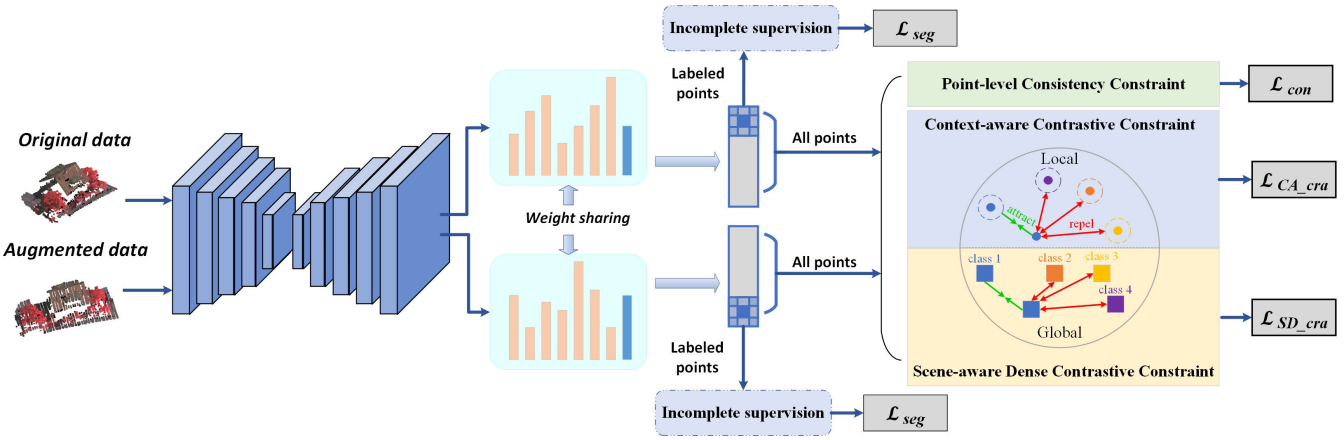


Fig. 1. Framework of DSDCL. DSDCL incorporates basic constraints, including a cross-entropy loss $\mathcal{L}_{\text{seg}}$ and a consistency loss $\mathcal{L}_{\text{con}}$, as well as two advanced contrastive constraints: the context-aware contrastive loss $\mathcal{L}_{\text{CD_cra}}$, which aligns anchor point features with the local semantic features of positive samples while distinguishing them from negative samples, and the scene-aware dense contrastive loss $\mathcal{L}_{\text{SD_cra}}$, which aligns the anchor's class prototype with the class prototype of positive samples while differentiating it from those of negative samples.

per-point jittering and the introduction of specific noise, results in the augmented point cloud $P' \in R^{(N\times 3)}$ as follows:

$$P' = s\mathbf{R}P + \mathbf{B} \tag{2}$$

$$\mathbf{R} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3}$$

$$\mathbf{B} = T + J \tag{4}$$

where s denotes the scale factor, $\mathbf{R}$ represents the 3-D rotation matrix parameterized by Euler angles, and the rotation angle θ around the z-axis belongs to the interval $[0, 2\pi]$. $\mathbf{B} \in R^{(N\times 3)}$ signifies the transformation matrix containing both global translations and point-wise jittering. Scaling and translation factors are uniformly selected from their corresponding intervals, with the scaling factor s, and the translation offset T. Point-wise jittering vectors are obtained by sampling from a Gaussian distribution. The point locations are jittered to obtain a noise displacement $J \in R^{(N\times 3)}$, where J follows a Gaussian distribution.

Moreover, enriching point cloud diversity by learnable soft masks that adaptively select the significant feature attributes. Specifically, the perturbed coordinate P' and original attribute P are concatenated as $\bar{P}$, then we calculate the attention score att of $\bar{P}$ which can be regarded as learnable transformations of input attribute weights. The final perturbed point cloud $\hat{P}$ is denoted by multiplying $\bar{P}$ with the normalized attention scores

$$\text{att} = \text{softmax}\left(\varphi\left(\bar{P}, \Theta\right)\right) \tag{5}$$

$$\hat{P} = \bar{P} \cdot \alpha\text{tt} \tag{6}$$

where $\varphi(\cdot)$ means a multilayer perceptron (MLP) with learnable parameters Θ and softmax$(\cdot)$ denotes the channel-normalized softmax function.

D. Point-Level Consistency Constraint

Existing methods perform point-level feature consistency by encouraging matching 3-D point pairs with robust and similar features in augmented transformations [23], [26], [38]. To enforce the constraints of the predictive consistency, we use the Jensen–Shannon divergence to constrain the probability distribution of each class between the original data branch and the perturbed data branch to minimize the prediction discrepancy between the two branches to the greatest extent possible. The point-level consistency constraint does not focus on the accuracy of category prediction across the two branches, but emphasizes ensuring the predictive consistency between the two branches. The mathematical formulation of the point-level consistency loss is delineated as follows:

$$\mathcal{L}_{\text{con}} = \frac{1}{2N}\sum_{i=1}^{N} \text{JS}\left(y_i \| \hat{y}_i\right) \tag{7}$$

where y_i and $\hat{y}_i$, respectively, represent the predicted probabilities for point p_i outputted by the original data branch P and the perturbed data branch $\hat{P}$. The term JS denotes the Jensen–Shannon divergence.

E. Context-Aware Contrastive Constraint

With only a minimal set of annotations available, merely emphasizing feature consistency is inadequate for grasping the complex structures characteristic of point clouds. The limitation poses challenges in accurately discerning both contextual differences and object details, thereby compromising the quality of segmentation outcomes. In addition, during the data collection process of ALS from air to ground, occlusions, and holes frequently occur. These issues significantly impact the integrity and quality of the point cloud data, thereby affecting the performance of models in learning features from the point cloud data. In response to these issues, we employ contrastive learning to enable the model to assimilate effective contextual structural information, thereby significantly enhancing the model's robustness to incomplete objects in environments characterized by occlusions or holes.

Specifically, we implement a context-aware contrastive constraint aimed at enhancing the model's ability to differentiate between similar data points by effectively leveraging the subtle differences in their contextual embeddings. Unlike traditional contrastive learning [39], we define context-aware contrastive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

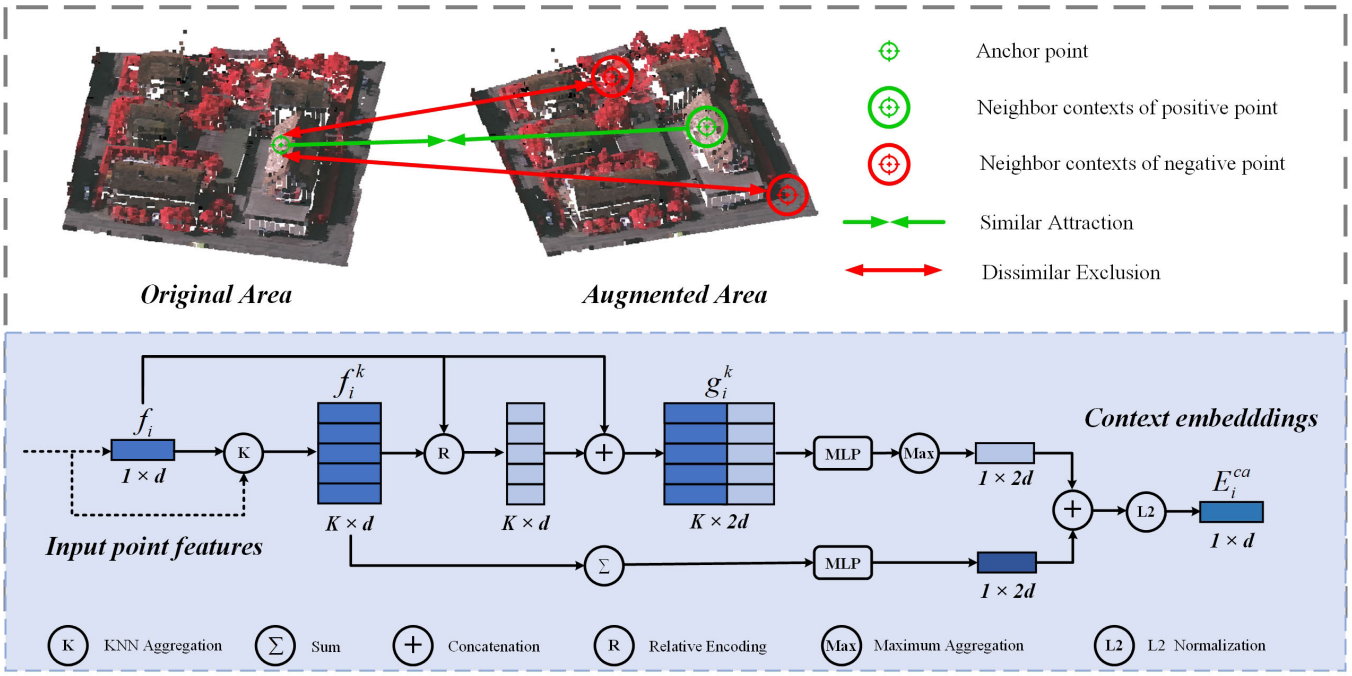


Fig. 2. Illustration of the context-aware contrastive constraint $\mathcal{L}_{\text{CA_cra}}$. Under this constraint, the features of green-marked anchor points in the original area are encouraged to be similar in contextual embeddings to those of the corresponding points (positive samples) in the augmented area, while being dissimilar to the contextual embeddings of other yellow-marked negative sample points. The blue box below illustrates the aggregation process of contextual feature embeddings.

constraint $\mathcal{L}_{\text{CA_cra}}$ to encapsulate this dynamic. This approach leverages the local semantic context of the anchor point to enhance contrastive learning at this stage, as illustrated in Fig. 2.

Within the designated feature set $F = \{f_1, \ldots, f_i, \ldots, f_N\}$, we establish a graph topology utilizing the feature graph. We then employ an MLP to compute the edge features. Subsequently, we aggregate these features using a maximum function. Finally, we calculate the $L2$ norm of the embedded features and perform normalization to enhance the feature representation

$$g_i^k = \left\{ f_i \oplus \left(f_i - f_i^k \right) \middle| f_i^k \in N(f_i) \right\} \tag{8}$$

$$h_i^k = \frac{1}{|\mathcal{N}(f_i)|} \sum_{f_i^k \in N(f_i)} f_i^k \tag{9}$$

$$\mathbf{E}_i^{\text{ca}} = \frac{\max\left(\mathcal{M}\left(g_i^k; \Phi\right)\right) \oplus \mathcal{M}\left(h_i^k; \Phi\right)}{\left\| \max\left(\mathcal{M}\left(g_i^k; \Phi\right)\right) \oplus \mathcal{M}\left(h_i^k; \Phi\right) \right\|_2} \tag{10}$$

where $N(f_i)$ denotes the features of the local neighborhood of the point p_i, constructed using the K-nearest neighbors based on Euclidean distance. Here, $\mathcal{M}$ denotes an MLP characterized by a set of learnable parameters Φ and the symbol $\oplus$ represents the concatenate operation, while $g_i^k, h_i^k \in R^{(K \times d)}$.

Then, we perform contrast constraint supervised prediction tasks, which entail a detailed classification for each point. Existing concepts of contrastive learning [27], [37] are centered around attracting a specific point, referred to as the anchor point p_i, toward its augmented point $\hat{p}_i$, while simultaneously distancing it from other points in the predictive space. The feature embedding of the anchor is compeled to be distinct from the feature embeddings of all its negative points p_j as shown in the following equation:

$$\mathcal{L}_{\text{cra}} = -\frac{1}{|\mathbf{M}_p|} \sum_{(p_i, p_j) \in \mathbf{M}_p} \log \frac{\exp\left(y_i \cdot \hat{y}_i / \tau\right)}{\sum_{(p_i, p_j) \in \mathbf{M}_p, i \neq j} \exp\left(y_i \cdot \hat{y}_j / \tau\right)} \tag{11}$$

where $\mathbf{M}_p$ is the set of indices of one-to-one matching point pairs randomly sampled from the point clouds of the original input $p_i \in P$ and its perturbed version $\hat{p}_i \in \hat{P}$. For the point pairs that match each other in point clouds P and $\hat{P}$, they are considered positive pairings, whose feature embeddings $(y_i, \hat{y}_i)$ are encouraged to be similar, while $(y_i, \hat{y}_j)$ are negative point pairs and encouraged to be different. τ represents a temperature hyperparameter.

Different from the above contrastive learning algorithm, we distribute the labeled points as anchor points in the graph topology, enforcing each anchor to be attracted toward the aggregated features of the neighborhood context of its corresponding perturbed points, while maintaining a distance from the neighborhood spaces of other points in the prediction space

$$\mathcal{L}_{\text{CA_cra}} = -\frac{1}{|\mathbf{M}_p|} \sum_{(i,j) \in \mathbf{M}_p} \mathcal{L}_{\text{ca_cra}}\left(y_i, \hat{\mathbf{E}}_i^{\text{ca}}\right) \tag{12}$$

$$\mathcal{L}_{\text{ca_cra}} = \log\left(\frac{\exp\left(y_i \cdot \hat{\mathbf{E}}_i^{\text{ca}} / \tau\right)}{\sum_{(i,j) \in \mathbf{M}_p, i \neq j} \exp\left(y_i \cdot \hat{\mathbf{E}}_j^{\text{ca}} / \tau\right)} \right) \tag{13}$$

where $\hat{\mathbf{E}}_i^{\text{ca}}$ and $\hat{\mathbf{E}}_j^{\text{ca}}$ represent the aggregated contextual features of the perturbed points $\hat{p}_i$ and $\hat{p}_j$, respectively.

For the features of the individual perturbed point, utilizing the subtle differences in the contextual embeddings of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

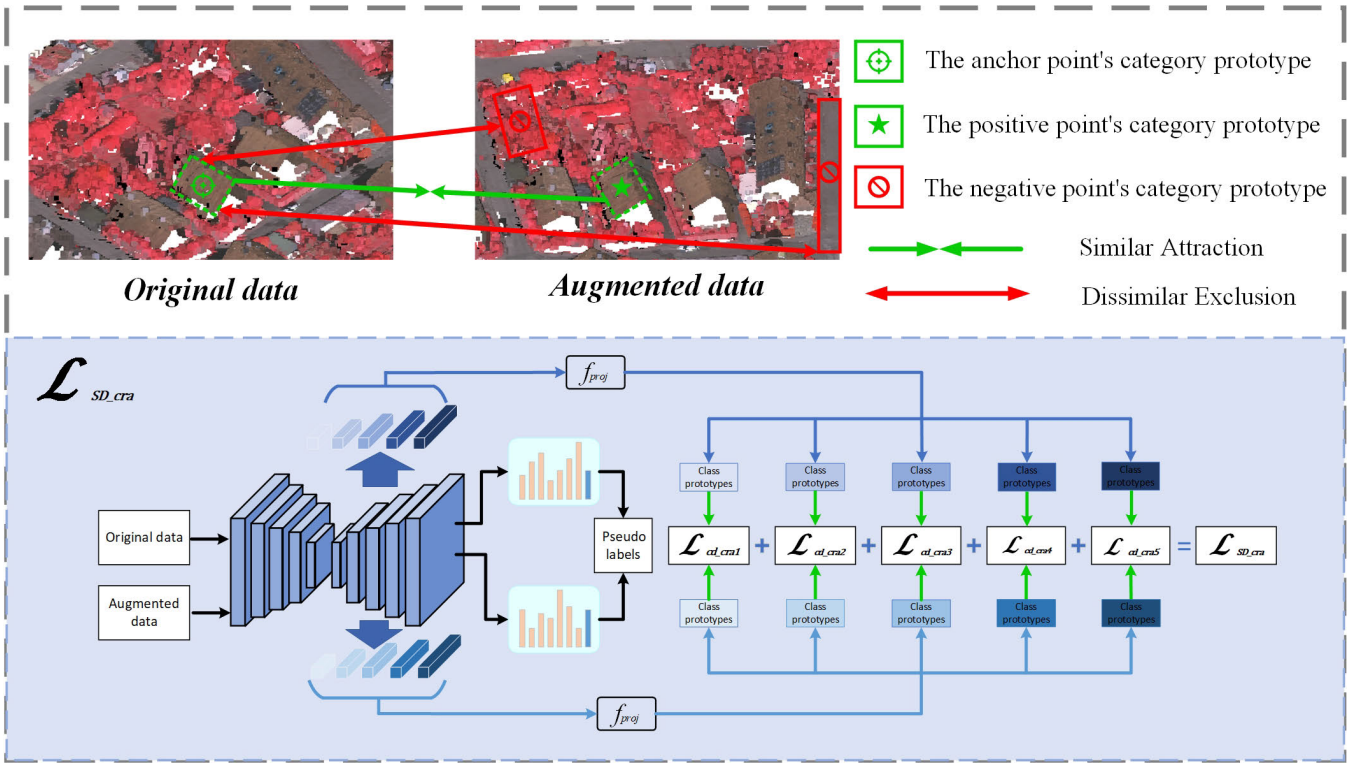


Fig. 3. Illustration of the scene-aware dense contrastive constraint $\mathcal{L}_{\mathrm{SD_cra}}$. Under this constraint, the category of the anchor point marked in green should be as consistent as possible with the category of the matched augmented point, while maintaining a distinction from the categories of other negative sample points marked in red. The blue box below demonstrates the process of extending category discrepancies to multilevel dense supervision.

anchor points to enhance current contrastive learning can more effectively improve the model's robustness to incomplete objects and its ability to differentiate similar data points. This approach will enhance the performance of point cloud feature learning in ALS scenarios characterized by occlusions and holes.

F. Scene-Aware Dense Contrastive Constraint

A semantic understanding of ALS point clouds with wide geographical coverage requires not only local information but also a comprehensive perception of the global scene level. Therefore, we propose a scene-aware dense contrastive learning through category distribution semantic differentiation and a multilevel dense supervision strategy.

In the network's training process, pseudo-labels for unlabeled points are generated from pairs of 3-D points originating from both the original data branch and the perturbed data branch. We dynamically supervise the training of the model using these class predictions as instructional signals. Specifically, for each data point p_i, the model assesses the probability S_i^c that it belongs to each category c. These probabilities are subsequently used to generate pseudo-labels Q_i for each point. Importantly, the selection of pseudo-labels is not arbitrary but is predicated on the confidence level of the model's predictions

$$Q_i = c, \quad \text{if } S_i^c > T_{\mathrm{pr}}, \ c \in [1, \ldots, C], \ i \in \left[1, \ldots, N_p\right] \tag{14}$$

where the pseudo-label Q_i is deemed reliable and suitable for subsequent training only if the predicted probability S_i^c for a category surpasses a predetermined threshold T_{pr}; pseudo-labels falling below this threshold T_{pr} are disregarded. C represents the total number of categories and N_p represents the number of points with pseudo-labels.

Subsequently, we utilize contrastive learning to achieve category discrepancies in the prototype features predicted by the pseudo-labels generated from the dual-branch data. We mitigate the adverse effects of erroneous pseudo-labels generated from individual samples by focusing on class prototype features, thereby enhancing the model's robustness to categorical distribution differences

$$L_{\mathrm{cd_cra}} = -\frac{1}{\left|M_p\right|} \sum_{(i,j)\in \mathbf{M}_p} \log \frac{\exp\left(\mathbf{E}_i^{\mathrm{pr}} \cdot \hat{\mathbf{E}}_i^{\mathrm{pr}} / \tau\right)}{\sum_{(i,j)\in \mathbf{M}_p, i\neq j} \exp\left(\mathbf{E}_i^{\mathrm{pr}} \cdot \mathbf{E}_j^{\mathrm{pr}} / \tau\right)} \tag{15}$$

where $\mathbf{E}_i^{\mathrm{pr}}$ and $\hat{\mathbf{E}}_i^{\mathrm{pr}}$, respectively, represent the aggregated prototype features of the categories to which point p_i and its perturbed counterpart $\hat{p}_i$ in the dual-branch data belong.

Then, we extend this categorical distribution disparity across multiple scales, pushing similar features from different scales more closely together. This multiscale approach is designed to finely tune the feature space within different layers of the network architecture, thereby fostering a closer alignment of features pertaining to identical categories. Concurrently, it aims to amplify the mutual information between the extracted features and their corresponding semantic labels, thereby bolstering the model's ability to discern and categorize distinct elements within the data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

This process, illustrated in Fig. 3, entails the independent generation of A_s at each scale s through a process of projecting the features F_s. This projection maintains the integrity and relevance of the features and labels across various scales, ensuring that the multiscale approach effectively captures the hierarchical structure of the data. The cumulative loss is then calculated as a weighted sum over these scales to form a scene-aware dense contrastive constraint $\mathcal{L}_{\text{SD_cra}}$

$$\mathcal{L}_{\text{SD_cra}} = w_s \sum_{s=1}^{S} \sum_{(i,j)\in M_{ps}} \mathcal{L}_{\text{cd_cra}}(i, j) \tag{16}$$

where the weights w_s are strategically assigned to each scale to optimize the model's sensitivity to features at different resolutions, thereby ensuring a balanced and comprehensive feature representation. The number of distinct scales S represents the breadth of the multiscale analysis.

As expounded in hybrid point cloud augmentation, the generation of dual-branch labeled points involves randomization, ensuring the independence of regions used in computing the stated loss at each scale. It ensures that the model's learning is not biased toward specific point cloud regions but is instead informed by a diverse and representative sample of the data's inherent features and labels across multiple scales. Through full-scale feature contrastive learning, this scaling serves to regulate the feature space across various network layers, encouraging proximity between features of the same class and maximizing the mutual information between features and their corresponding semantic labels.

IV. Experiment

In order to comprehensively evaluate the effectiveness of DSDCL, this study conducted extensive experiments on three publicly available ALS point cloud reference datasets. We adopted a combination of quantitative and qualitative evaluation methods, comparing various performance indicators and conducting data visualization analysis to gain a deeper understanding of DSDCL's performance in various aspects.

A. Experimental Datasets

In order to evaluate and analyze, the study selected three ALS point cloud datasets in densely populated urban areas for semantic annotation: ISPRS dataset [40], DFC2019 dataset [41], and LASDU dataset [42].

1) ISPRS Dataset: This dataset consists of the Leica ALS 50 system and an infrared camera used to obtain auxiliary color information. ALS point cloud data and aerial images were collected in the Stuttgart region of Germany, with a density range of 4–7 points/m^2. The dataset comprises nine categories: low vegetation, power lines, concrete ground, trees, cars, roofs, fences, exterior walls, and shrubs. For training and testing purposes, the dataset was divided into two parts, with the training set containing 753 859 points and the testing set containing 411 721 points. There is a significant variation in the number of points among different categories, with a focus on low vegetation, roofs, concrete ground, and trees. The dataset is subsampled using 0.45-m grid sizes, and during both training and testing phases, labels for deleted points were assigned based on nearest neighbor points.

2) LASDU Dataset: This dataset was obtained through the Leica ALS70 system installed on airplanes and studied an urban area located in the Heihe River Basin in northwest China, with an average point density ranging from 3 to 4 points/m^2. The entire region is divided into four contiguous parts, with two parts used as training sets and the other two parts as testing sets. The LASDU dataset has a total of approximately 3.12 million points, with a relatively low point cloud density. Research typically uses the raw data for training and testing. The dataset predefines five categories: ground, artifacts, low vegetation, trees, and buildings.

3) DFC2019 Dataset: This dataset was acquired in two cities in the United States, Omaha in Nebraska and Jacksonville in Florida, covering an area of approximately 100 km^2 and containing around 200 million points. The nominal pulse spacing during point cloud acquisition is 0.8 m. In addition to XYZ coordinates, point intensity, and return count information are also provided. The dataset is divided into five semantic classes: high vegetation, ground, buildings, water bodies, and bridges. Following the data partitioning by Zeng et al. [43], 100 and 10 slices are, respectively, taken as training and testing data.

B. Implementation Details

The configuration details are explained here. Initially, due to the large volume of the raw data, we performed subsampling beforehand to enhance computational efficiency while retaining the structural details of the point cloud. In the training process, we set the grid size to 0.45 m for the ISPRS dataset, 0.72 m for the DFC2019 dataset, and 0.2 for the LASDU dataset. The parameters λ_1 and λ_2 in Formula (1) were set to 0.5 during the experiments. Due to the nonfixed number of points in each subcloud, batch size is not constant and the maximum total limit for each training step is 12 000 points during the training process. The total number of epochs was fixed at 100, each consisting of 500 iteration steps. We use RandLA-Net [11] as the base network framework, along with an Adam optimizer initialized with a learning rate of 10^{-3}. The models were built using the TensorFlow framework, running on an Intel[1] Xeon[1] Gold 6330 CPU with a clock speed of 2.00 GHz, and leveraging an NVIDIA GeForce RTX3090 GPU.

C. Evaluation Metrics

To assess the segmentation outcomes, we employed overall accuracy (OA) and the average $F1$ score (Avg. $F1$) as metrics to measure the segmentation accuracy of the ALS point clouds. The $F1$ score takes into account both recall and precision of the segmentation model, making it more appropriate where there is an uneven distribution of categories. In addition, the OA typically signifies the overall segmentation accuracy across all categories, calculated as the percentage of correctly classified points among the total test points. For each category, OA and $F1$ score are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

[1]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

TABLE I

SEMANTIC SEGMENTATION RESULTS ON ISPRS DATASET. THE RESULTS INCLUDE THE $F1$ SCORE (%) OF EACH CATEGORY, THE AVG.$F1$ SCORE (%), AND THE OAS (%)

Setting	Method	F1 Score									Avg. F1	OA
		Power	Low veg.	Imp. Surf.	Car	Fence	Roof	Facade	Shrub	Tree		
Full Sup.	PointNet++ [45]	57.9	79.6	90.6	66.1	31.5	91.6	54.3	41.6	77	65.6	81.2
	PointSIFT [46]	55.7	80.7	90.9	77.8	30.5	92.5	56.9	44.4	79.6	67.7	82.2
	D-FCN [47]	70.4	80.2	91.4	78.1	37.0	93	60.5	46.0	79.4	70.7	82.2
	RandLANet [11]	69.0	80.1	91.4	73.1	36.2	93.7	58.4	45.8	82.7	70.1	83.1
	DANCE-NET [48]	68.4	81.6	**92.8**	77.2	38.6	93.9	60.2	47.2	81.4	71.2	83.9
	KPConv [10]	74.2	81.7	91.8	**80.0**	29.6	**94.7**	63.3	44.7	80.6	71.2	83.8
	PointCNN [49]	61.5	**82.7**	91.8	75.8	35.9	92.7	57.8	49.1	78.1	69.5	83.3
	VD-LAB [50]	69.3	80.5	90.4	79.4	38.3	89.5	59.7	47.5	77.2	70.2	81.4
	RFFS-Net [51]	75.5	80.0	90.5	78.5	**45.5**	92.7	57.9	48.3	75.7	71.6	82.1
	LGENet [52]	**76.5**	82.1	91.8	80.0	40.6	93.8	**64.7**	**49.9**	**83.6**	**73.7**	**84.5**
Weak Sup. (1%)	Baseline [11]	56.8	**78.2**	**88.9**	67.4	24.2	**94.0**	58.4	41.9	**82.0**	65.8	80.6
	PSD [23]	58.3	74.2	85.3	53.2	13.5	92.1	51.7	**44.8**	79.2	61.4	79.1
	SQN [44]	31.1	77.9	87.3	66.0	26.1	93.5	55.1	39.2	80.3	61.8	80.8
	DR-Net [53]	35.4	72.3	84.9	67.6	31.1	91.3	56.3	40.3	79.3	62.1	77.5
	DSDCL	**73.0**	76.8	88.0	**78.2**	**39.7**	92.6	**61.8**	38.9	81.9	**70.1**	**81.8**
Weak Sup. (1‰)	Baseline [11]	37.8	**76.0**	**87.2**	3.6	6.4	**92.3**	47.0	35.3	**78.6**	51.6	78.6
	PSD [23]	46.9	74.1	81.4	4.7	10.8	90.6	44.6	34.7	75.8	51.5	75.5
	SQN [44]	32.6	71.3	83.4	33.4	8.9	87.2	45.3	36.3	70.7	52.1	74.5
	DR-Net [53]	17.4	60.4	75.1	46.4	20.1	85.7	52.8	37.9	70.5	51.8	67.0
	DSDCL	**58.5**	72.3	86.5	**64.1**	**34.2**	91.0	**56.7**	**50.8**	77.2	**65.7**	**79.7**

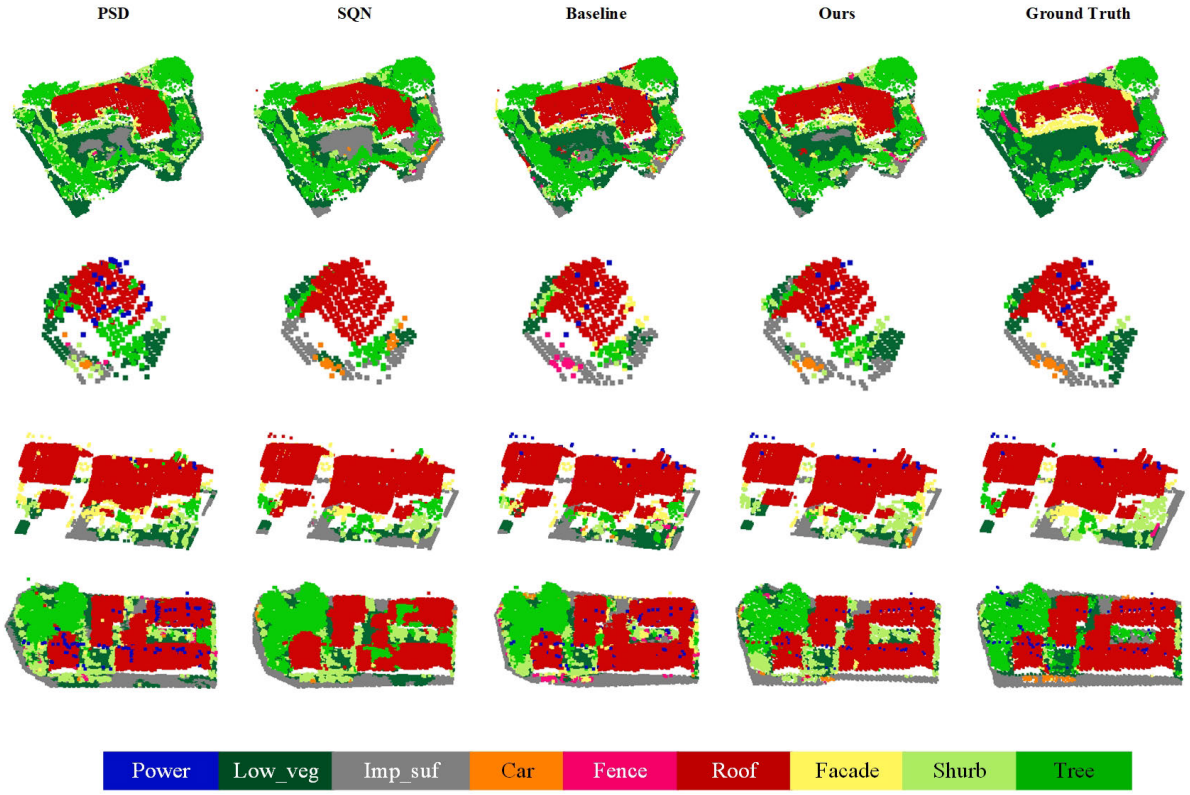


Fig. 4. Semantic segmentation results for the ISPRS dataset.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{18}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{19}$$

$$F1 = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{20}$$

where true positives (TPs) are the correctly identified positive predictions, true negatives (TNs) are the correctly identified negative predictions, false positives (FPs) are the incorrectly identified positive predictions, and false negatives (FNs) are the incorrectly identified negative predictions.

D. Evaluation Results on ISPRS Dataset

In Fig. 4, we exhibit the segmentation results garnered from the ISPRS dataset, providing an insightful qualitative evaluation comparing the segmentation efficacy of PSD [23], SQN [44], Baseline [11], and DSDCL. A noteworthy observation from the figure is the challenge presented by the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SEMANTIC SEGMENTATION RESULTS ON DFC19 DATASET. THE RESULTS INCLUDE THE $F1$ SCORE (%) OF EACH CATEGORY, THE AVG.$F1$ SCORE (%), AND THE OAS (%)

Setting	Method	F1 Score					Avg. F1	OA
		Ground	Hig_veg	Building	Water	Bridge		
Full Sup.	PointNet++ [45]	98.3	95.8	79.7	4.4	7.3	57.1	92.7
	PointCNN [49]	98.7	97.2	84.9	44.1	65.3	78	93.8
	PointConv [54]	97.3	95.8	**93.6**	74.5	69.2	**86.1**	95.3
	DGCNN [55]	97.9	93.2	90.4	**88.2**	54.4	84.2	95.1
	KPConv [10]	98.4	94.2	87.4	43.0	77.5	80.1	94.5
	D-FCN [47]	99.1	98.1	89.9	45.0	73.0	81	95.6
	DANCE-NET [48]	99.1	93.9	87.0	58.3	**83.9**	84.4	96.8
	LGENet [52]	**99.3**	**98.3**	92.8	47.4	79.1	83.4	**98.4**
Weak Sup. (1%)	Baseline [11]	95.0	94.0	87.3	28.4	50.4	71.0	93.7
	PSD [23]	95.6	94.3	88.9	48.0	47.7	74.9	94.4
	SQN [44]	**97.8**	95.9	89.4	43.8	44.6	74.3	95.8
	DR-Net [53]	97.5	**96.0**	**94.7**	**72.1**	71.9	**86.4**	95.7
	DSDCL	95.8	94.2	90.1	55.6	**75.8**	82.3	**96.3**
Weak Sup. (0.1%)	Baseline [11]	93.1	92.7	80.4	30.1	28.0	64.9	90.7
	PSD [23]	92.9	93.9	85.0	30.5	31.8	66.8	90.9
	SQN [44]	**97.0**	94.4	88.2	34.2	34.1	69.6	**94.7**
	DR-Net [53]	94.3	**95.5**	**92.1**	50.6	58.6	78.2	91.6
	DSDCL	95.2	93.0	88.5	**50.6**	**70.7**	**79.6**	94.6

nuanced differences in geometrical features among urban elements, such as the subtle distinctions between the small sample categories of cars and fences, and the complexity in segregating vegetative components like shrubs and trees. Moreover, the complexity is compounded when considering the segmentation of slender objects such as power lines, which require a high level of precision. For instance, the discernment of DSDCL in segregating the yellow building façades from the green trees showcases a significant advancement over the other methods. This is particularly evident in the first row, where DSDCL accurately demarcates the red roof edges, avoiding the confusion between yellow façades with green shrubs and trees observed in PSD and SQN. In the selected region of the second row, which contains a sparse distribution of low vegetation, power lines, cars, roofs, and trees, DSDCL exhibits an exceptional segmentation capability that consistently maintains closer fidelity to the ground truth amidst the confluence of multiple categories within a limited small area. This is a notable improvement over the baseline method that tends to merge cars with impervious surfaces. In addition, DSDCL exhibits commendable performance in delineating subtle power areas (highlighted in blue), as shown in the third row, where other methods struggle with the powers' proximity to other structures. Similarly, the oversight of power lines is also apparent in the comparative algorithms' results in the fourth row, where the other methods fail to accurately segment these features. Moreover, the outcomes in the fourth row illustrate that our algorithm yields a notable improvement in segmenting sparse car samples located at the boundaries, compared to the baseline method, which erroneously classifies these few car samples as fences.

Table I presents the experimental results of various comparison methods within the domains of full and weak supervision. In the condition of weak supervision with only 1% of labels, the comparison with the baseline demonstrates that DSDCL significantly enhances the OA and $F1$ scores across all categories, highlighting its ability to improve model performance using limited labels. Specifically, it surpasses PSD by 8.7% and 2.7% in OA and $F1$ scores, respectively, exceeds SQN by 8.3% and 1%, and outperforms the baseline by 4.3% and 1.2%, respectively. It is noteworthy that under the same evaluation criteria, DSDCL achieves the same average $F1$ valve as the fully supervised RandLA-Net. Specifically, in the realm of individual categories such as power, fence, car, and facade, DSDCL achieves the highest $F1$ score, thereby underscoring its robust feature extraction capability across different urban landscape objects. Under extremely sparse annotation conditions with only 0.1% of labeled data, although the performance of most comparative algorithms significantly decreases for categories with few samples such as power lines, cars, and fences, DSDCL exhibits the best performance in these three categories, achieving $F1$ scores of 58.5, 64.1, and 34.2, respectively. Specifically, compared with the latest MLP-based method (DR-Net), DSDCL increases OA by 12.7% and Avg.$F1$ by 13.9%. This striking contrast underscores the effectiveness of DSDCL in leveraging dense supervision context-aware contrastive constraints to address the challenges posed by label scarcity on model learning dynamics and its eventual classification capabilities.

E. Evaluation Results on DFC19 Dataset

Fig. 5 illustrates the segmentation results on the DFC19 dataset, showcasing the complexities encountered during the segmentation process, particularly in distinguishing closely arranged and visually similar urban elements, such as the boundaries between water bodies and adjacent land, or the segmentation of bridges from their surroundings. The segmentation of these elements becomes increasingly challenging due to their proximity and subtle variations when observed from an aerial perspective. Fig. 5 (first row) provides a perspective

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

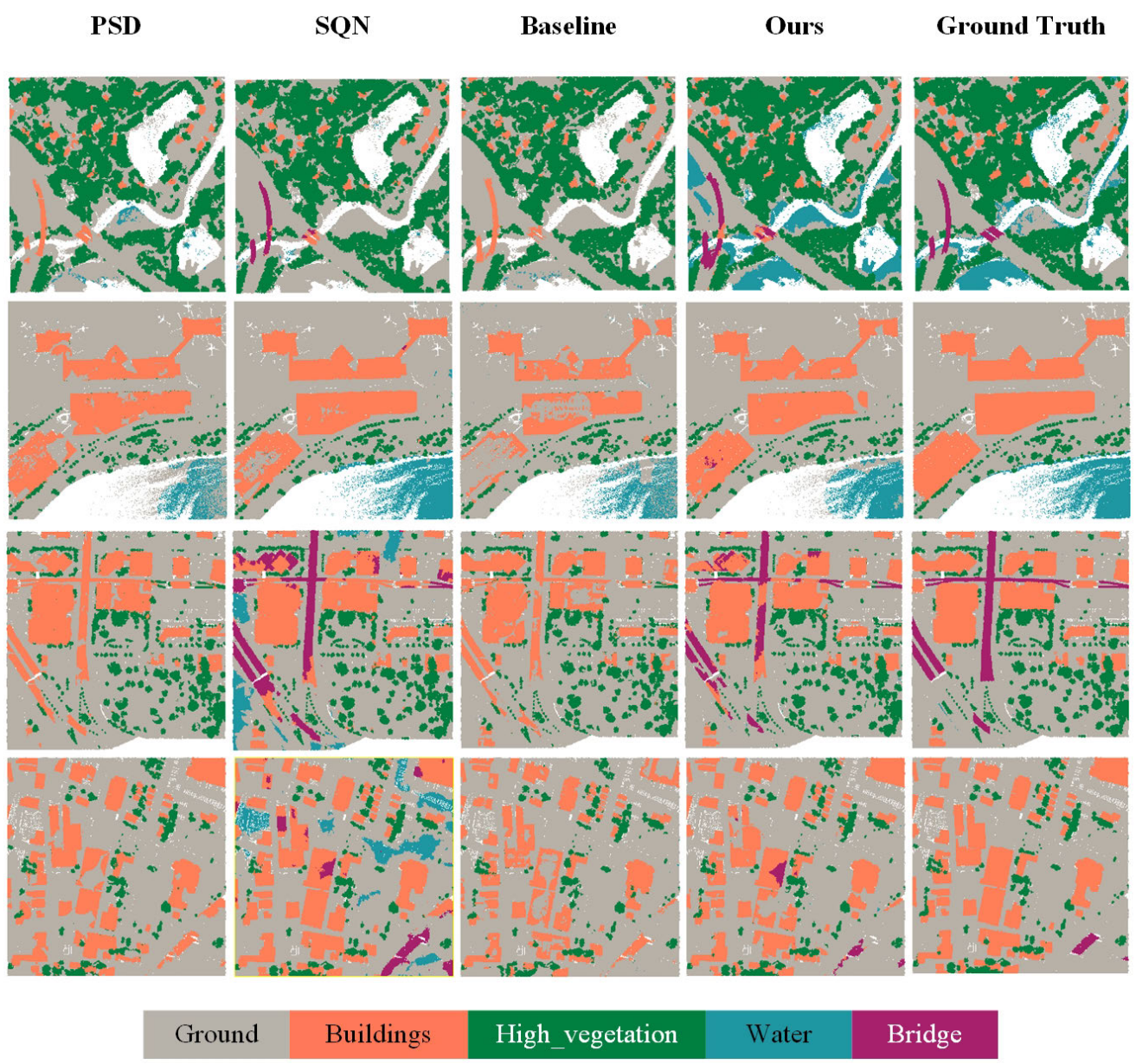


Fig. 5. Experimental results for the DFC19 dataset.

on transportation networks, showcasing DSDCL's advanced performance in segmenting bridges (purple) from water bodies (blue). Here, the PSD and SQN methods reveal significant segmentation errors, especially in delineating the intricate structure of the bridges, while DSDCL delivers a segmentation quality that is consistent with the ground truth, indicating a nuanced comprehension of the distinctive geometrical features. Proceeding to the second row, DSDCL continues to demonstrate its superior capability in accurately identifying and separating buildings from the ground, a task in which the baseline and PSD methods exhibit less distinction, leading to blurred boundaries and compromised structural detail. The third row presents a challenging urban scenario characterized by the overlapping of multiple complex categories, including bridges, buildings, vegetation, and ground. DSDCL adeptly distinguishes the bridges and adjacent buildings, whereas the baseline method erroneously introduces the water category into this segmentation results. In the bottom row, DSDCL precisely segments high vegetation (green) from buildings (orange) and ground (gray), contrary to PSD and SQN which exhibit notable misclassification at the interface of these classes. DSDCL maintains a clear delineation that echoes the fidelity of the ground truth. In each scenario, DSDCL demonstrates a remarkable proficiency in segmenting diverse urban elements with a high degree of precision, effectively managing the challenges posed by the similarities in appearance and proximity of different classes. This highlights the powerful learning ability of the DSDCL method for complex environments, successfully capturing the key details required for large-scale urban landscape segmentation.

The experimental outcomes delineated in Table II provide a comprehensive comparison across various methods under

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

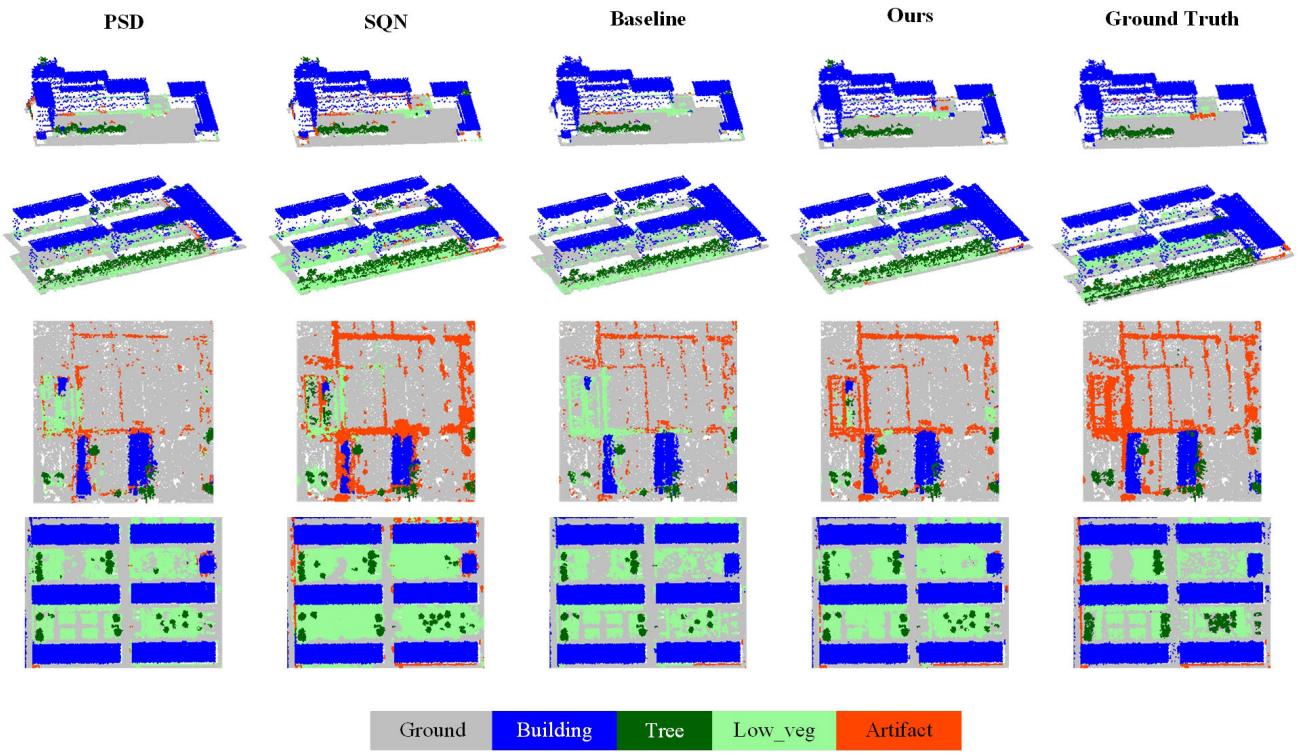


Fig. 6. Experimental results for the LASDU dataset.

TABLE III

SEMANTIC SEGMENTATION RESULTS ON LASDU DATASET. THE RESULTS INCLUDE THE $F1$ SCORE (%) OF EACH CATEGORY, THE AVG.$F1$ SCORE (%), AND THE OAS (%)

Setting	Method	F1 Score					Avg. F1	OA
		Artifacts	Building	Ground	Low veg	Tree		
Full Sup.	PointNet++ [45]	87.7	90.6	82.0	63.2	31.3	71.0	82.8
	PointCNN [49]	89.3	92.8	84.1	62.8	31.7	72.1	85.0
	PointConv [54]	89.6	94.3	84.6	67.5	36.4	74.5	85.9
	DGCNN [55]	90.5	93.2	81.6	63.3	37.1	73.1	85.5
	ShellNet [56]	89.5	92.8	84.6	**68.5**	35.7	74.2	85.6
	PosPool [57]	88.3	93.7	83.9	61.0	38.3	73.0	83.5
	KPConv [10]	89.9	96.1	**85.1**	56.9	33.6	72.3	85.9
	RRDAN [43]	**91.6**	**96.6**	84.1	66.3	**48.3**	**77.4**	**87.7**
Weak Sup. (1%)	Baseline [11]	91.7	95.9	86.1	58.5	44.2	75.3	86.3
	PSD [23]	91.7	96.0	87.4	57.0	41.6	74.7	86.7
	SQN [44]	90.3	95.9	87.4	60.7	43.8	75.6	85.1
	DR-Net [53]	87.7	94.9	83.0	**61.8**	41.2	73.7	83.2
	DSDCL	**91.9**	**96.0**	**88.4**	58.7	**50.0**	**77.0**	**87.3**
Weak Sup. (0.1%)	Baseline [11]	**91.5**	93.7	86.2	53.1	26.5	70.2	84.8
	PSD [23]	90.7	95.0	86.7	52.3	26.7	70.3	85.1
	SQN [44]	85.0	95.5	**87.5**	54.9	**40.8**	72.7	81.2
	DR-Net [53]	84.5	94.0	83.7	58.6	30.4	70.2	80.4
	DSDCL	90.9	**95.7**	86.1	**56.4**	37.9	**73.4**	**85.4**

both full and weak supervision settings, utilizing the diverse landscapes of the dataset. Remarkably, under the 1% labeling framework, DSDCL yields a 7.4% improvement over PSD, 8.0% over SQN, and an 11.3% improvement over the baseline. particularly excelling in building, water and bridge categories with $F1$ scores of 90.1%, 55.6%, and 75.8%, respectively. In the 0.1% weak supervision setting, DSDCL does not experience significant reductions in $F1$ scores across most categories. It is worth noting that although DSDCL's average $F1$ score is 4.1% lower than DR-Net under the 1% setting, primarily because of DR-Net's superior segmentation performance on the water category, DSDCL's OA is 1.4% higher. Moreover, under the 0.1% setting, both OA and average $F1$ scores surpass DR-Net by 3.0% and 1.4%, respectively. The higher OA can be attributed to DSDCL's superior generalization capability across multiple categories,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

particularly in challenging environments, leading to a more consistent performance across diverse categories. For instance, it achieves 95.2% for ground, 93% for high vegetation, 88.5% for building, and 50.6% for Water, showcasing its robustness and effectiveness in maintaining performance even with extremely sparse labeling. When compared with algorithms under fully supervised conditions, the performance gap in the $F1$ scores for individual categories between our method and top-performing methods like KPConv and D-FCN is notably narrow. This suggests that our weakly supervised approach is adept at extracting deep, meaningful features that are on par with those derived from fully supervised methods. Despite the commendable OA achieved by methods such as DANCE-NET and LGENet, DSDCL still maintains a competitive edge in terms of average $F1$ scores, particularly in categories that are traditionally challenging such as water and bridge. When positioned against fully supervised competitive popular algorithms, our method demonstrates commendable performance, closely rivaling and in some instances surpassing, methodologies like RandLA-Net and D-FCN in specific categories.

F. Evaluation Results on LASDU Dataset

Fig. 6 illustrates the segmentation results on the LASDU dataset, highlighting the challenges in distinguishing various urban elements, such as separating low vegetation from tree canopies and accurately classifying buildings within other structures. In the first and second rows of Fig. 6, we observe independent sections of the urban landscape primarily comprising building structures. It is evident from the boundaries between the ground (gray) and vegetation (green), as well as a small number of artifacts (red), that our method showcases a distinct improvement over the baseline and PSD methods, which tend to confuse the artifacts with lower vegetation. In the third row, focusing on an urban park setting, DSDCL exhibits superior performance in distinguishing between ground and artifacts (red), an area where the baseline and PSD methods struggle, often misclassifying artifacts as ground or vice versa. The distinction made by DSDCL is sharp and adheres more faithfully to the ground Truth. The fourth row emphasizes the segmentation of low vegetation (light green) from ground (gray), a task in which DSDCL excels by maintaining the integrity of the low vegetation class, whereas the baseline and SQN methods exhibit a tendency to incorrectly merge it with the ground category.

The analytical insights derived from Table III present the performances exhibited by a variety of methods under both full and weak supervision frameworks. Under the 1% weak supervision paradigm, DSDCL achieves significant improvements in average $F1$ scores, outperforming PSD, SQN, the baseline, and DR-Net by 2.3%, 1.4%, 1.7%, and 3.3%, respectively. This superior performance is notably evident in the artifacts, building, water, and tree categories, where DSDCL attains $F1$ scores of 91.9%, 96.0%, 88.4%, and 50.0%, respectively, showcasing DSDCL's exceptional capability to effectively manage sparse labels. Even under the extremely sparse 0.1% annotation condition, despite facing inherent challenges, DSDCL maintains commendable $F1$ scores of 95.7% and 56.4% for the building and low vegetation categories, respectively, surpassing other methods. This stark contrast highlights DSDCL's capability to leverage contrastive learning to mitigate the impact of label scarcity on learning efficacy and classification accuracy. It is noteworthy that under the 1% supervision condition, DSDCL not only surpasses the fully supervised KPConv in terms of $F1$ and OA scores, improving by 4.7% and 1.4%, respectively. Even under the extremely sparse 0.1% annotation condition, DSDCL still outperforms fully supervised KPConv by 1.1% in terms of $F1$ score, highlighting the exceptional efficacy and robustness of DSDCL in leveraging minimal labels to achieve superior performance.

TABLE IV
ABLATION STUDIES OF DIFFERENT COMBINATIONS OF THE LOSSES

Group	Module			ISPRS		DFC19		LASDU	
	Base.	$\mathcal{L}_{CA_cra}$	$\mathcal{L}_{SD_cra}$	Avg. F1	OA	Avg. F1	OA	Avg. F1	OA
1	✓			56.8	74.4	70.2	93.6	71.4	83.7
2	✓	✓		60.5	76.1	73.6	93.8	72.9	84.3
3	✓		✓	59.8	77.0	74.6	94.0	73.1	84.5
4	✓	✓	✓	65.7	79.7	79.6	94.6	73.4	85.4

V. DISCUSSION

This section evaluates the effectiveness of different combinations of loss functions and analyzes the improvement of each individual constraint. Subsequently, a complexity analysis of the model was conducted.

A. Ablation Study

In this section, we discuss the performance of each module in the DSDCL framework and present the results of ablation experiments in Table IV, based on experiments conducted with 0.1% labels from the ISPRS, DFC19, and LASDU datasets.

1) Context-Aware Contrastive Constraint: Group 1 is the basic loss with added cross-entropy loss and consistency constraint. We add the context-aware contrastive constraint $\mathcal{L}_{\text{CA_cra}}$ to the network and define the model as group 2 in the network. From the comparison between groups 3 and 2 at 0.1% setting, group 3 outperforms group 2 the average $F1$ score by 3.7% and OA by 1.7% for ISPRS. In DFC19's metrics, group 3 improved by 3.4% in the average $F1$ score and 0.2% in OA. For LASDU, the increases are 1.5% for the average $F1$ score and 0.6% for OA. These results indicate that context-aware contrastive constraints can further improve model performance, as it utilizes adjacent context to enhance the model's feature learning ability.

2) Scene-Aware Dense Contrastive Constraint: We also tested the performance of adding individual scene-aware dense contrastive constraint $\mathcal{L}_{\text{SD_cra}}$ to the network. In the ISPRS dataset, there is an increase of approximately 3.0% in the average $F1$ score and an enhancement of 2.6% in OA. For the DFC19 dataset, the average $F1$ score improves by 4.4%, and the OA shows an increase of 0.4%. In the LASDU dataset, the average $F1$ score has increased by 1.7%, and the OA has improved by 0.8%. The results demonstrate that $\mathcal{L}_{\text{SD_cra}}$ can effectively enhance the segmentation performance of a weakly supervised framework under limited labeling.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

TABLE V
COMPARISON OF MODEL COMPLEXITY ON DIFFERENT DATASETS

Methods	Parameters (M)	Training Time (h)			Inference Time (s)		
		ISPRS	DFC19	LASDU	ISPRS	DFC19	LASDU
Baseline [11]	4.99	0.78	1.35	1.33	31.65	11.92	24.47
PSD [23]	5.03	0.76	1.35	1.31	31.50	12.80	23.65
SQN [44]	3.48	0.92	1.38	1.35	35.44	15.19	26.01
DSDCL	5.08	1.21	2.12	2.08	32.62	12.08	24.16

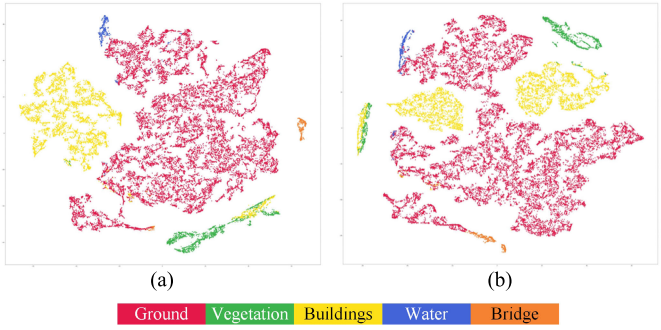


Fig. 7. Comparison of t-SNE visualization.

B. Model Complexity Analysis

In Table V, we compare the complexity of baseline, PSD, and SQN models by utilizing the number of network parameters, model training time, and inference speed. Since ours only integrates a lightweight context-aware module on top of the baseline network for contrastive learning, it adds only the parameters of the CA module compared to the baseline. Hence, ours has only 0.09M more parameters than the baseline. However, due to the involvement of more calculations in the contrastive learning process, our algorithm has slightly improved training time compared to the baseline. Inference time refers to the testing phase, which requires the model to test 106 points of the dataset. Based on the table data, it can be observed that the test times of ours and other comparison algorithms are basically similar.

C. Visualization of High-Dimensional Features

To further analyze the performance of DSDCL, we transform the high-dimensional features obtained from this method into 2-D spatial distributions using t-SNE [58] to visualize the structure and patterns of the high-dimensional features and understand the clusters, outliers, and patterns in the segmentation results. As shown in Fig. 7, the point-level representations of segmentation results use different colors to indicate different semantic categories. An analysis of the cluster shapes across these categories reveals that DSDCL achieves markedly more concentrated cluster groups compared to PSD. Specifically, the intracategory point density is significantly higher. This compactness implies that within the original high-dimensional space, distances between points belonging to the same category are notably shorter, whereas the intercategory distances are considerably expanded. This spatial arrangement underscores the effectiveness of contrastive learning in enhancing the similarity amongst points within the same category and augmenting the distinction between categories, thereby fostering a more discernible high-dimensional data representation.

VI. CONCLUSION

In this work, we propose a DSDCL for ALS point cloud weakly supervised semantic segmentation, fully utilizing sparse annotations by considering semantic neighborhood relations in scenes and the global features of 3-D scenes. We introduce a context-aware contrastive constraint to enhance the model's ability to differentiate similar data points by leveraging subtle contextual differences, improving feature learning robustness in environments with incomplete objects. In addition, we utilize scene-aware dense contrastive learning throughout multiple layers of the model, directly supervising the feature spaces in the initial convolutional and attention layers. Conceptually, we regard the network as a function that transforms the input point cloud into a multiscale representation, spread throughout its various stages, and imbue it with class-aware constraints in the feature space. This contrastive constraint enhances category discrepancies in prototype features from dual-branch pseudo-labels, improving model robustness to categorical distribution differences and mitigating the impact of erroneous labels. Thorough experiments with three benchmark ALS datasets show that DSDCL markedly surpasses existing weakly supervised approaches, and in certain instances, it even surpasses the performance of various fully supervised learning models.

REFERENCES

[1] C. Alexander, P. K. Bøcher, L. Arge, and J.-C. Svenning, "Regional-scale mapping of tree cover, height and main phenological tree types using airborne laser scanning data," *Remote Sens. Environ.*, vol. 147, pp. 156–172, May 2014.

[2] Y. Wang et al., "International benchmarking of the individual tree detection methods for modeling 3-D canopy structure for silviculture and forest ecology using airborne laser scanning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 9, pp. 5011–5027, Sep. 2016.

[3] L. Matikainen, K. Karila, P. Litkey, E. Ahokas, and J. Hyyppä, "Combining single photon and multispectral airborne laser scanning for land cover classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 164, pp. 200–216, Jun. 2020.

[4] L. Matikainen, K. Karila, J. Hyyppä, P. Litkey, E. Puttonen, and E. Ahokas, "Object-based analysis of multispectral airborne laser scanner data for land cover classification and map updating," *ISPRS J. Photogramm. Remote Sens.*, vol. 128, pp. 298–313, Jun. 2017.

[5] Y. Zhou, R. Huang, T. Jiang, Z. Dong, and B. Yang, "Highway alignments extraction and 3D modeling from airborne laser scanning point clouds," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 102, Oct. 2021, Art. no. 102429.

[6] H. Song and J. Jung, "An unsupervised, open-source workflow for 2D and 3D building mapping from airborne LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 6067–6084, 2024.

[7] L. Fang, J. Liu, Y. Pan, Z. Ye, and X. Tong, "Semantic supported urban change detection using ALS point clouds," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 118, Apr. 2023, Art. no. 103271.

[8] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and X. Gao, "DCENet: Diff-feature contrast enhancement network for semi-supervised hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511514.

[9] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," 2016, *arXiv:1612.00593*.

[10] H. Thomas, C. R. Qi, J.-E. Deschaud, B. Marcotegui, F. Goulette, and L. J. Guibas, "KPConv: Flexible and deformable convolution for point clouds," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6411–6420.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

[11] Q. Hu et al., "RandLA-Net: Efficient semantic segmentation of large-scale point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11108–11117.

[12] H. Zhao, L. Jiang, J. Jia, P. Torr, and V. Koltun, "Point transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 16259–16268.

[13] Z. Zeng, H. Qiu, J. Zhou, Z. Dong, J. Xiao, and B. Li, "PointNAT: Large-scale point cloud semantic segmentation via neighbor aggregation with transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5704618.

[14] Z. Luo, Z. Zeng, W. Tang, J. Wan, Z. Xie, and Y. Xu, "Dense dual-branch cross attention network for semantic segmentation of large-scale point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5700216.

[15] Z. Luo et al., "D2T-net: A dual-domain transformer network exploiting spatial and channel dimensions for semantic segmentation of urban mobile laser scanning point clouds," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 132, Aug. 2024, Art. no. 104039.

[16] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508614.

[17] T. Guo, L. He, F. Luo, X. Gong, L. Zhang, and X. Gao, "Learnable background endmember with subspace representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5501513.

[18] T. Guo, L. He, F. Luo, X. Gong, Y. Li, and L. Zhang, "Anomaly detection of hyperspectral image with hierarchical antinoise mutual-incoherence-induced low-rank representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510213.

[19] F. Luo, Y. Liu, Y. Duan, T. Guo, L. Zhang, and B. Du, "SDST: Self-supervised double-structure transformer for hyperspectral images clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511614.

[20] T. Zeng, F. Luo, T. Guo, X. Gong, W. Shu, and Y. Zhao, "HSCN: Semi-supervised ALS point cloud semantic segmentation via hybrid structure constraint network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5708513.

[21] Y. Zhang, Z. Li, Y. Xie, Y. Qu, C. Li, and T. Mei, "Weakly supervised semantic segmentation for large-scale point cloud," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 4, pp. 3421–3429.

[22] X. Xu and G. H. Lee, "Weakly supervised semantic point cloud segmentation: Towards 10x fewer labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 13706–13715.

[23] Y. Zhang, Y. Qu, Y. Xie, Z. Li, S. Zheng, and C. Li, "Perturbed self-distillation: Weakly supervised large-scale point cloud semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 15520–15528.

[24] J. Wei, G. Lin, K.-H. Yap, F. Liu, and T.-Y. Hung, "Dense supervision propagation for weakly supervised semantic segmentation on 3D point clouds," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 6, pp. 4367–4377, Jun. 2024.

[25] J. Hou, B. Graham, M. Nießner, and S. Xie, "Exploring data-efficient 3D scene understanding with contrastive scene contexts," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15587–15597.

[26] M. Li et al., "HybridCR: Weakly-supervised 3D point cloud semantic segmentation via hybrid contrastive regularization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14930–14939.

[27] S. Xie, J. Gu, D. Guo, C. R. Qi, L. Guibas, and O. Litany, "PointContrast: Unsupervised pre-training for 3D point cloud understanding," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*. Glasgow, U.K.: Springer, 2020, pp. 574–591.

[28] X. Zhao et al., "Contrastive learning for label efficient semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10623–10633.

[29] T. Wang and P. Isola, "Understanding contrastive representation learning through alignment and uniformity on the hypersphere," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 9929–9939.

[30] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[31] X. Wang, R. Zhang, C. Shen, T. Kong, and L. Li, "Dense contrastive learning for self-supervised visual pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3024–3033.

[32] P. Wang and W. Yao, "A new weakly supervised approach for ALS point cloud semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 188, pp. 237–254, Jun. 2022.

[33] Y. Lin, G. Vosselman, and M. Y. Yang, "Weakly supervised semantic segmentation of airborne laser scanning point clouds," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 79–100, May 2022.

[34] P. Wang, W. Yao, and J. Shao, "One class one click: Quasi scene-level weakly supervised point cloud semantic segmentation with active learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 89–104, Oct. 2023.

[35] J. Zhao et al., "SemanticFlow: Semantic segmentation of sequential LiDAR point clouds from sparse frame annotations," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5701611.

[36] Y. Su, M. Cheng, Z. Yuan, W. Liu, W. Zeng, and C. Wang, "Multi-stage scene-level constraints for large-scale point cloud weakly supervised semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5705118.

[37] L. Jiang et al., "Guided point contrastive learning for semi-supervised point cloud semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 6423–6432.

[38] C.-K. Yang, J.-J. Wu, K.-S. Chen, Y.-Y. Chuang, and Y.-Y. Lin, "An MIL-derived transformer for weakly supervised point cloud segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11830–11839.

[39] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.

[40] J. Niemeyer, F. Rottensteiner, and U. Soergel, "Contextual classification of LiDAR data and building object detection in urban areas," *ISPRS J. Photogramm. Remote Sens.*, vol. 87, pp. 152–165, Jan. 2014.

[41] M. Bosch, K. Foster, G. Christie, S. Wang, G. D. Hager, and M. Brown, "Semantic stereo for incidental satellite images," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2019, pp. 1524–1532.

[42] Z. Ye et al., "LASDU: A large-scale aerial LiDAR dataset for semantic labeling in dense urban areas," *ISPRS Int. J. Geo-Inf.*, vol. 9, no. 7, p. 450, Jul. 2020.

[43] T. Zeng, F. Luo, T. Guo, X. Gong, J. Xue, and H. Li, "Recurrent residual dual attention network for airborne laser scanning point cloud semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5702614.

[44] Q. Hu et al., "SQN: Weakly-supervised semantic segmentation of large-scale 3D point clouds," in *Proc. 17th Eur. Conf. Comput. Vis. (ECCV)*. Tel Aviv, Israel: Springer-Verlag, 2022, pp. 600–619, doi: 10.1007/978-3-031-19812-0_35.

[45] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, "PointNet++: Deep hierarchical feature learning on point sets in a metric space," 2017, *arXiv:1706.02413*.

[46] M. Jiang, Y. Wu, T. Zhao, Z. Zhao, and C. Lu, "PointSIFT: A SIFT-like network module for 3D point cloud semantic segmentation," 2018, *arXiv:1807.00652*.

[47] C. Wen, L. Yang, X. Li, L. Peng, and T. Chi, "Directionally constrained fully convolutional neural network for airborne LiDAR point cloud classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 50–62, Apr. 2020.

[48] X. Li, L. Wang, M. Wang, C. Wen, and Y. Fang, "DANCE-NET: Density-aware convolution networks with context encoding for airborne LiDAR point cloud classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 128–139, Aug. 2020.

[49] Y. Li, R. Bu, M. Sun, W. Wu, X. Di, and B. Chen, "PointCNN: Convolution on X-transformed points," in *Proc. 32nd Int. Conf. Neural Inf. Process. Syst.* Red Hook, NY, USA: Curran Associates, 2018, pp. 828–838.

[50] J. Li et al., "VD-LAB: A view-decoupled network with local–global aggregation bridge for airborne laser scanning point cloud classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 186, pp. 19–33, Apr. 2022.

[51] Y. Mao et al., "Beyond single receptive field: A receptive field fusion-and-stratification network for airborne laser scanning point cloud classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 188, pp. 45–61, Jun. 2022.

[52] Y. Lin, G. Vosselman, Y. Cao, and M. Y. Yang, "Local and global encoder network for semantic segmentation of airborne laser scanning point clouds," *ISPRS J. Photogramm. Remote Sens.*, vol. 176, pp. 151–168, Jun. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.

[53] L. Zhang and Y. Bi, "Weakly-supervised point cloud semantic segmentation based on dilated region," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5105020.
[54] W. Wu, Z. Qi, and L. Fuxin, "PointConv: Deep convolutional networks on 3D point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9621–9630.
[55] Y. Wang, Y. Sun, Z. Liu, S. E. Sarma, M. M. Bronstein, and J. M. Solomon, "Dynamic graph CNN for learning on point clouds," *ACM Trans. Graph.*, vol. 38, no. 5, pp. 1–12, 2019.
[56] Z. Zhang, B. Hua, and S. Yeung, "ShellNet: Efficient point cloud convolutional neural networks using concentric shells statistics," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1607–1616.
[57] Z. Liu, H. Hu, Y. Cao, Z. Zhang, and X. Tong, "A closer look at local aggregation operators in point cloud analysis," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*. Glasgow, U.K.: Springer, 2020, pp. 326–342.
[58] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.



Ziwei Luo received the B.S. degree in spatial information and digital technology and the Ph.D. degree in surveying science and technology from China University of Geosciences (Wuhan), Wuhan, China, in 2018 and 2024, respectively.

She is currently an Assistant Professor at the School of Computer and Artificial Intelligence, Wuhan Textile University, Wuhan. Her research interests include point cloud semantic segmentation, point cloud analysis and processing, and 3-D scene understanding.



Tao Zeng received the B.E. and M.E. degrees in mechanical engineering and automation from Southwest Petroleum University, Chengdu, China, in 2011 and 2015, respectively, and the Ph.D. degree in mechatronics engineering from Xi'an Technological University, Xi'an, China, in 2024.

He is currently a Post-Doctoral Researcher at the School of Electronic Information Engineering, Sichuan University, Chengdu, and Chengdu Qianjia Technology Company, Chengdu. His research interests include point cloud data processing.



Xinyi Jiang received the B.E. degree in software engineering from Jiangxi Normal University, Nanchang, China, in 2023. She is currently pursuing the M.E. degree with China University of Geosciences (Wuhan), Wuhan, China.

Her research interests include deep learning, point cloud processing, and semantic segmentation.



Qingyu Peng received the B.E. degree in marine engineering from Wuhan University of Technology, Wuhan, China, in 2020. He is currently pursuing the Ph.D. degree with Huazhong University of Science and Technology, Wuhan.

His research interests include robot machining and robot measurement.



Ying Ma received the M.E. degree in surveying and mapping engineering from China University of Geosciences (Wuhan), Wuhan, China, in 2022, where she is currently pursuing the Ph.D. degree in surveying and mapping science and technology.

Her research interests include landslide geohazard chain identification and mechanism research.



Zhong Xie received the B.S., M.S., and Ph.D. degrees in cartography and geographic information engineering from China University of Geosciences (Wuhan), Wuhan, China, in 1990, 1998, and 2002, respectively.

He is currently a Professor with the School of Geography and Information Engineering, China University of Geosciences (Wuhan). His research interests include deep learning, 3-D rebuilding and spatial analysis, and image processing.



Xiong Pan received the Ph.D. degree in geodesy and surveying engineering from Wuhan University, Wuhan, China, in 2005.

He is currently a Professor with the School of Computer and Artificial Intelligence, Wuhan Textile University, Wuhan. His main interests include data mining and satellite navigation positioning technology.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:03:48 UTC from IEEE Xplore. Restrictions apply.